Journal of Network and Computer Applications 242 (2025) 104251



Contents lists available at ScienceDirect

Journal of Network and Computer Applications

journal homepage: www.elsevier.com/locate/jnca



Research paper



Effective ensemble learning phishing detection system using hybrid feature selection

Aaron Connolly [a], Hany F. Atlam [a,b,*]

[a] Cyber Security Centre, WMG, University of Warwick, Coventry, CV4 7AL, United Kingdom
[b] Computer Science and Engineering Department, Faculty of Electronic Engineering, Menoufia University, Menouf, 32952, Egypt

ARTICLE INFO

Keywords:
Phishing email detection
Ensemble learning
Hybrid feature selection
Machine Learning
Stacking approach

ABSTRACT

Phishing emails pose a significant threat to individuals and os, with traditional detection methods struggling to keep pace with the evolving sophistication of these attacks. Conventional Machine Learning (ML) approaches exhibit several limitations in achieving satisfactory accuracy levels when challenged with the evolving sophistication of phishing techniques. To effectively mitigate this challenge, the implementation of an advanced detection system incorporating innovative and sophisticated ML algorithms is crucial. Therefore, this paper proposed a novel stacking ensemble learning approach that leverages hybrid feature selection. The proposed model enhances the effectiveness of phishing detection by combining predictions from multiple ML algorithms, each utilising different subsets of features extracted from various parts of the email, including the header, body, and URLs. This comprehensive feature set ensures that the model captures a wide range of characteristics that differentiate phishing emails from legitimate ones. Extensive experiments were conducted to evaluate the effectiveness of the proposed model. The experimental results demonstrate that the proposed model achieves an impressive accuracy of 99.53% and an F1-measure of 0.9955, surpassing the highest accuracy of 99.10% obtained by any individual ML algorithm and outperforming the most effective phishing detection systems documented in recent literature. This significant improvement in accuracy highlights the efficacy of ensemble learning in this domain. Furthermore, the increase in accuracy is achieved with only a minimal 1.6 ms increase in detection time, making the model practical for real-world applications. This paper contributes significantly to the field of phishing detection by demonstrating the effectiveness of ensemble learning techniques in combination with hybrid feature selection. The proposed model offers a practical and effective solution to the problem of phishing, with the potential to significantly reduce the number of malicious emails reaching users' inboxes.

1. Introduction

Phishing is a social engineering attack targeting computer users with electronic messages delivered most commonly through email 1u (Khonji et al., 2013). Phishing aims to trick the user into downloading malicious software or disclosing sensitive information (Salloum et al., 2021). Threat actors often impersonate a reputable body, organisation or person to increase the chance of gaining the users trust (Bountakas et al., 2021). Phishing is especially ubiquitous in the modern threat landscape, and is a challenge shared by organisations of all sizes across all industries worldwide. Despite the long-standing prevalence of phishing attacks, phishing has been identified as the most common cyber threat facing both individuals and organisations. Cloudflare's 'Phishing Threats Report' states an estimated up to 90% of successful cyber attacks start with email phishing (Dzuba and Cash, 2023).

In attempts to tackle the phishing problem, researchers have deployed Machine Learning (ML) detection systems. These systems aim to identify patterns and characteristics shared by phishing emails, enabling the detection of unseen phishing campaigns (El Naqa and Murphy, 2015). ML techniques have significantly enhanced the accuracy of phishing detection when compared to traditional red/green list-based approaches, which fail to identify zero-day phishing attacks (Khonji et al., 2013). Although ML approaches are significantly more effective than list-based techniques, the accuracy achieved in the previous studies can be improved upon with innovative detection techniques that are underutilised in the existing works, such as ensemble learning.

The stark increase in the number of phishing attacks is alarming and poses a significant threat to both individuals and organisations (Salloum et al., 2021). Despite advancements in detection techniques,

* Corresponding author at: Cyber Security Centre, WMG, University of Warwick, Coventry, CV4 7AL, United Kingdom.
E-mail address: hany.atlam@warwick.ac.uk (H.F. Atlam).

https://doi.org/10.1016/j.jnca.2025.104251
Received 25 June 2024; Received in revised form 8 June 2025; Accepted 18 June 2025
Available online 30 June 2025
1084-8045/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

phishing tactics present an ongoing challenge. With the vast quantity of fraudulent emails sent across the internet each day, even a slight improvement in accuracy would result in significant volumes of malicious emails being blocked. Although existing solutions are capable of achieving very good accuracy, they fail to detect approximately 1000 phishing emails out of every 100,000 (Bountakas and Xenakis, 2023). In a large organisation, encountering such a considerable volume of phishing emails is realistic. Even 1% of phishing emails arriving in the mailboxes of users pose a substantial security risk. This becomes especially dangerous when malicious emails are delivered to the inboxes of users with little security awareness (Dewis and Viana, 2022). Through proper training and organisational cybersecurity awareness, the risk of a user falling victim to a phishing email is reduced. However, even those with cybersecurity experience can be deceived by phishing. It is imperative that the most effective email filtering techniques are implemented to reduce the number of malicious emails arriving in user mailboxes as much as possible (Khonji et al., 2013).

To tackle this problem, we propose a novel stacking ensemble learning approach designed to enhance phishing email detection. The proposed system leverages hybrid feature extraction techniques to combine predictions from multiple ML algorithms, thus improving detection accuracy. The proposed approach integrates a two-tiered stacking approach: the first tier (base learners) consists of the two best-performing algorithms, each utilising different subsets of features for classification, while the second tier (meta learner) takes the predictions of these base learners to make the final classification. The ensemble learning technique employed is built upon hybrid features extracted from various parts of the email, including the header, body, and URLs. This comprehensive feature set ensures that the model captures a wide range of characteristics that differentiate phishing emails from legitimate ones. The proposed system is designed to maintain high accuracy without significantly increasing detection time. To evaluate the performance of the proposed stacking ensemble model, extensive experiments were conducted. The results demonstrate that our model achieves an accuracy of 99.53% and an F1-measure of 0.9955. These results surpass the highest accuracy of 99.10% obtained by any individual machine learning algorithm, highlighting the efficacy of ensemble learning in this domain. Additionally, the increase in accuracy is achieved with only a minimal 1.6 ms increase in detection time, highlighting the model's potential for real-world application.

The main contributions of this research are summarised as follows:

- Introducing a novel hybrid feature selection that combines diverse algorithms to offer an accurate and comprehensive representation for detecting phishing emails.
- Proposing a novel stacking ensemble learning phishing detection system utilising a two-tiered stack, enhancing detection accuracy through an innovative multi-layer approach.
- Evaluating the effectiveness of the proposed stacking technique through detailed experimental results, comparing it against individual machine learning algorithms and the most effective phishing detection systems in the literature.
- Demonstrating that the combination of stacking ensemble learning with hybrid features significantly enhances the detection effectiveness of phishing emails.

The remainder of this paper is structured as follows. Section 2 presents related works in the literature where state-of-the-art phishing detection systems are discussed and critically evaluated. Section 3 introduces the proposed ensemble model and discusses the techniques utilised in their design and implementation. Section 4 presents the experimental resuls and discuss the evaluation of the proposed approach against state-of-the-art systems from the literature.Finally Section 5 provides a conclusion to the paper.

Table 1
List of abbreviations

Abbreviation	Expanded abbreviation
ML	Machine Learning
NN	Neural Network
FPR	False Positive Rate
FNR	False Negative Rate
CNN	Convolutional Neural Network
NLP	Natural Language Processing
LDA	Latent Dirichlet Allocation
NB	Naive Bayes
RF	Random Forest
KNN	K-Nearest Neighbours
DT	Decision Tree
LR	Logistic Regression
SVM	Support Vector Machines
SVC	Support Vector Classifier
SMO	Sequential Minimal Optimization
LSTM	Long Short-Term Memory
BOW	Bag-Of-Words
TF-IDF	Term Frequency- Inverse Document Frequency
CS-SVM	Cuckoo Search Support Vector Machine
RBF	Radial Basis Function
SVD	Singular Value Decomposition
NMF	Non-negative Matrix Factorisation
KM-SMOTE	K-Means Synthetic Minority Oversampling Technique
GNB	Gaussian Naive Bayes
BNB	Bernoulli Naive Bayes
JS	JavaScript
ROC	Receiver Operating Characteristic
AUC	Area Under Curve

2. Related work

Researchers have deployed a variety of ML techniques to tackle the phishing problem. Smadi et al. (2018) propose a reinforcement learning-based Neural Network (NN), utilising 50 features from email headers and body content, resulting in an accuracy of 98.63%. This study utilised PhishingCorups for phishing emails, SpamAssassin for ham and spam emails and PhishTank for URLs, resulting in a final dataset of 7,315 phish and 4951 ham emails. Although achieving a high accuracy and respectable FPR, using an unbalanced dataset containing approximately 35% more phishing emails than legitimate emails can create a bias. Furthermore, the high number of features can result in a decreased detection time and requires a larger sample size in order to effectively train the model.

An approach utilising the combination of Keras word embedding and a Convolutional Neural Network (CNN) is proposed by Hiransha et al. (2018). Multiple datasets are deployed in the study, with one containing emails with headers and another containing emails without headers. The dataset including the email header included 4,082 legitimate and 501 phishing emails whereas the dataset without header information included 5,088 legitimate and 612 phishing emails. For testing- each set was given approximately 4250 emails from the IWSPA-AP Shared Task committee. The best accuracy achieved by the model was 96.8%- using the emails without a header. However, the dataset used in this study for training is not open source and therefore the model cannot be accurately compared against other studies. In addition, the paper did not test their models on newer emails to evaluate whether they can capture the evolution of phishing email attacks.

A NN is deployed to achieve a high accuracy of 99.85% with a FPR of 0.045 by Fang et al. (2019) in the model THEMIS, utilising the Jose Nazario dataset for phishing emails and sourcing legitimate emails from WikiLeaks, Enron and SpamAssassin. However, the use of an imbalanced dataset with a small sample of 1,000 phishing emails can introduce a classification bias in the model. Similarly, Alotaibi et al. (2020) employed CNN to achieve a 99.42% accuracy making use of a dataset consisting of 7,315 phishing emails from PhishingCorpus and 6,057 emails from SpamAssassin- including 4,150 ham emails and

1,897 spam emails. However, as features are only extracted from the email body and subject, the header is not considered in the classification. Additionally, utilising solely language-based features results in key features such as URLs being excluded from classification. A common weakness shared by nearly all NN models is that they do not perform well in explainability, where NNs are complex and similar to a black box, making understanding the classification process very difficult (Zhang et al., 2022).

Another avenue explored in the literature is the implementation of Natural Language Processing (NLP). Ramanathan and Wechsler (2013) proposed a unique approach with the detection of impersonation of a named entity (people, organisations and locations). NLP is combined with AdaBoost in order to classify the emails. Conditional random field is employed to extract the named entities from the email and Latent Dirichlet Allocation (LDA) is used to extract topics from the text. The dataset is built from several sources, including 4,150 legitimate emails from SpamAssassin, 4,550 phishing emails from PhishingCorpus and 2,200 phishing emails from SPAM Archive. The model resulted with an F1-measure of 96% and the impersonation detection provided approximately 88.1% discovery rate of impersonated entities. Similarly, a URL level detection system is proposed by Sahingoz et al. (2019), in which URLs are analysed using NLP features and word vectors. The study deploys 7 different ML algorithms- Naive Bayes (NB), Random Forest (RF), K-Nearest Neighbours (KNN), AdaBoost, K-star, Sequential Minimal Optimization (SMO) and DT. The dataset used in the study was created by the authors and published as the Ebbu2017 phishing dataset- including 36,400 legitimate URLs and 37,175 phishing URLs. The best performance was achieved with the RF classifier- reaching an accuracy of 97.98%. However, the authors identified that the model struggles to correctly classify short URLs with no paths.

Gualberto et al. (2020) proposed a model utilising NLP in two methods- bag-of-words (BoW) with Term Frequency-Inverse Document Frequency (TF-IDF), and LDA. Different machine learning classification algorithms were implemented, including- Support Vector Machines (SVM), NB, Logistic Regression (LR), KNN, DT, RF, XGBoost and Multilayer Perceptron (MLP). The best performing model utilised the combination of LDA and XGBoost achieving a validation accuracy of 99.95%. PhishingCorpus was used to source email samples for this study, providing 2,279 phishing emails. This was used alongside 4,150 ham emails from SpamAssassin PublicCorpus. However, the study did not consider the evolution of phishing emails, as the deployed phishing emails used were outdated. In the same way, Bountakas et al. (2021) provide a comparison of 3 different NLP techniques for phishing detection using 5 different machine learning algorithms. The authors tested the models on 2 different datasets- a balanced dataset including 1,407 benign emails from the Enron email corpus and 1,407 phishing emails from the Nazario dataset, and an imbalanced dataset including 14,000 legitimate and 1,407 phishing emails from the same sources. The method which achieved the best results was the Word2Vec NLP combined with a RF classifier resulting in a 98.95% accuracy with a FPR of 1.4%. However, the study addresses phishing email detection as a text classification task- ignoring valuable information in the email headers and URLs. Phish Responder is proposed by Dewis and Viana (2022) using NLP and deep learning for phishing and spam detection. The model is used on different datasets using Long Short-Term Memory (LSTM) for textual data sourced from the SpamAssassin dataset achieving 99% accuracy and MLP for numerical data achieving a 94% accuracy from an online Kaggle email spam dataset. However, the system does not detect in real time. Furthermore, the Kaggle dataset is not standardised or deployed in other studies.

The literature shows a variety of traditional ML approaches such as the heuristic approach for detecting phishing at the URL level, proposed by Nguyen et al. (2014) where 6 heuristics were extracted from the URL, including a reputation analysis. The URLs used in this study are sourced from PhishTank including a total of 11,660 phishing websites. An additional 10,000 sites are split into 10 training datasets including either 1,000 phish or 1,000 legitimate sites. The model achieved a detection rate of 97.16%. However, this could be improved through the implementation of more heuristic parameters. Additionally, as identified by the authors, the combination with another technique such as fuzzy or NN could increase the detection rate. Also, Egozi and Verma (2018) propose a system in which 26 semantic features were extracted from emails. The most effective model tested was a weighted DT, resulting in the correct classification of 80% of the phishing emails and 95% of the legitimate emails. However, the results of this study show a low detection rate for phishing emails, which can be attributed to the feature selection technique and the use of an imbalanced dataset including only 14% phishing emails.

Furthermore, SVM is employed by Niu et al. (2017) through the use of Cuckoo Search Support Vector Machine (CS-SVM) utilising the Radial Basis Function (RBF) in classifying emails based on 23 features extracted from the header, body and URLs. The dataset used includes 1,387 phishing emails from the Nazario and MillerSmiles datasets and 20,071 ham emails from the Enron dataset. However, there is a potential bias in the imbalanced dataset used along with a small number of phishing emails in the sample.

Smadi et al. (2015) proposed a unique approach in which data mining is employed to extract data patterns alongside 23 hybrid features drawn from previous studies. The classifiers used were RF, J48 and PART, with RF achieving the highest accuracy at 98.87%. The dataset contains 4559 phishing emails from the Nazario dataset, and a further 4559 legitimate emails sampled from SpamAssassin. However, the majority of the phishing email samples used in the study were dated between 2004 and 2007, which does not represent modern phishing threats. The implementation of a fuzzy system is demonstrated by Che et al. (2017) utilising different levels of phishing indicators- very legitimate, legitimate, suspicious, phishing and very phishing. It aims to assist users with a supervised approach to help recognise phishing- as opposed to blocking the emails. However, the study does not provide evaluation data to display the effectiveness of the approach.

Nb and Kp (2018) investigated different ML algorithms for phishing email detection through the implementation of TF-IDF followed by Singular Value Decomposition (SVD) for feature extraction and Non-negative Matrix Factorisation (NMF) for dimensionality reduction. The dataset used includes 10,283 emails in the training, and 8,495 emails in the testing set where the emails were sourced from the IWSPA-AP Shared Task email dataset. The author extracted 30 features and fed them into different ML algorithms. The highest accuracy in training was achieved with DT and RF, however, the testing accuracy is not provided due to the overfitting of the model. The authors argued this issue due to the high imbalance in the dataset. Furthermore, M. and Pais (2022) proposed a system in which 4 header features are extracted and tokenised then input into lemmatisation. Different vectorisation algorithms are used with FastText-CBOW holding the best performance. Different ML algorithms were deployed, in which RF achieved the highest accuracy with an impressive 99.50% outperforming DT, SVM, XGB and LR algorithms. The dataset was large including 6,295 phishing emails from the phishing corpus and 9,135 legitimate emails from HamCorpus. However, with a small number of features that are only extracted from the header, the body content and textual information are missed. Furthermore, an attacker using a compromised account to send emails from a legitimate sender will bypass half of the features used in the study.

Hybrid feature extraction techniques are first presented in the literature by Ma et al. (2009), where the authors proposed a 7 feature design utilising the WestPac dataset, consisting of 659,673 emails. The sample contains 613,048 legitimate emails and 46,525 phishing emails. The dataset contains significant imbalance, with phishing samples only comprising 7% of the data. Multiple different classifiers were tested with DT achieving the highest performance utilising 10-fold cross-validation. The model performs well in validation at 99.8% accuracy. However, testing on unseen data would offer a greater indication of

the model's capacity to generalise. Also, this study evaluated only on accuracy which is an inappropriate metric for imbalanced data and the results might be biased towards the majority class. Another approach combining a hybrid between several feature types was proposed by Rahmi et al. (2011), where 7 features were extracted from both the email header and the email body and the classification was calculated with the Bayesian network algorithm achieving an accuracy of 96% with a FPR and FNR of 4%. The samples used are from the Nazario and SpamAssassin datasets, including 4,559 phishing emails and 2,365 legitimate emails. However, the study did not consider the evolution of phishing emails, as the deployed phishing emails used were outdated ranging from 2004–2007.

PhishNet-NLP is a hybrid model proposed by Verma et al. (2012) extracting information from the header, body and URLs. Each section is assigned equal importance through the implementation of a majority vote in the classification process. Standard features were extracted from the header and the URLs. For the body, NLP techniques are deployed to extract keywords in the text. The model is deployed with a set of 2,000 phishing emails collected from the Nazario dataset, alongside a further 1,000 legitimate emails. However, a weighing of importance between the 3 feature categories would result in a more granular system. Furthermore, a multi-stage hybrid approach is used by Daeef et al. (2016) using a hybrid selection of features from the email body, header and URL totalling 48 features. The dataset used includes 4,450 phishing emails from the Nazario dataset and 4,400 emails from the SpamAssassin dataset. The final validation accuracy of the proposed approach was 99.91% utilising the RF classifier. However, due to the large number of features; the processing time for each email takes a long time, around 90 s which is too long for real-time processing. Reducing the number of features by removing the least significant features could decrease dimensionality and the overall classification time.

Another hybrid system is proposed by Yang et al. (2019) in which 9 features were derived from each email, encompassing five structural features and four behavioural features. The model uses SVM for classification and was implemented with a set of 500 legitimate and 500 phishing emails. The phishing emails were sampled from the Nazario dataset and the legitimate emails were from the SpamAssassin dataset. This method resulted in an accuracy of 97.50% and a FPR of 4.5%. However, the resulting accuracy is low, which can be attributed to the small sample set.

Yadav and Panda (2021) extract phishing terms from the email body and subject using TF-IDF to select textual features. Additionally, 5 features were extracted with HTML parsing. The feature set was reduced with a dimensionality reduction algorithm leaving 15 features combining HTML and textual features. Several ML algorithms are tested including LR, J45 and RF. The best accuracy of 99% was gained with RF. The dataset used in the study includes 4,559 phishing emails from Nazario and a further 4,559 from the SpamAssassin dataset. However, the study utilises a small number of features compared to other studies from the literature.

A model to detect spear-phishing emails is proposed by Ding et al. (2021). A highly imbalanced dataset of 417 spear-phishing emails, 5,562 benign emails and 8,404 common phishing emails is used. A data enhancement algorithm K-Means Synthetic Minority Oversampling Technique (KM-SMOTE) was used to address the dataset imbalance. 21 Stylometric, 3 forwarding and 3 reputation features were extracted from the emails. When classifying spear-phishing emails, an F1-score of 97.16% was achieved with RF. The approach used is unique, utilising features from the body and additionally the reputation of the sender. However, the dataset used is not comparable with those used in standard phishing detection systems and reputation features are ineffective against zero-day attacks from senders without a negative reputation.

Recently, ensemble learning methods have been employed to increase the accuracy of existing models. Anandita Yadav et al. (2017) introduced an ensemble learning method in which 15 features were extracted from the email header and body and 5 ML algorithms were combined to classify the emails. These algorithms were Gaussian Naive Bayes (GNB), Bernoulli Naive Bayes (BNB), RF, KNN and SVM. The combinations of the different algorithms increased the accuracy from 94.1% to 98.02% with ensemble learning. However, the sample size for the dataset was small, with only 500 phishes and 2,550 legitimate emails from SpamAssassin being used. As the dataset was small and imbalanced, it does not provide a full indication of the effectiveness of the proposed technique. Similarly, Kumar et al. (2020) proposed an ensemble approach combining a probabilistic NN and SVM, achieving 98.3% accuracy. However, the use of a small, unstandardised dataset consisting of 1,291 ham emails and 404 phishing emails limits the comparability of the results with those of other studies.

Another ensemble learning model for the detection of phishing at the URL level using canopy feature selection was proposed by Karim et al. (2023). The study compares the effectiveness of different ML algorithms, with the ensemble model combining LR, Support Vector Classifier (SVC) and DT resulting in an accuracy of 98.12%. The dataset contained 11,054 URLs from a Kaggle repository. However, the distribution between phishing links and legitimate links lacks specific elaboration, and the dataset's comparability with those utilised in other studies remains undisclosed. Also, the dataset's validity and the limited elaboration on the selected features pose challenges in terms of comparability with existing literature. Furthermore, Bountakas and Xenakis (2023) proposed a model designed to fuse the content and textual traits of the email with the extraction of 22 features. The features are divided into the categories; body, syntactic, header and URL features. The study combines two ML algorithms for the final classification using a DT for content-based features and KNN for text-based features. Both stacking and soft voting ensemble learning were tested, with soft voting achieving the best performance with an accuracy of 99.43%. The dataset used included 32,051 benign emails from the Enron and SpamAssassin datasets and 3,460 phishing emails sourced from the Nazario dataset, including 1,472 recent emails from 2015–2020 and the remaining emails from before this range. This study shows great promise in the use of ensemble learning techniques for phishing email classification. A summary of the machine learning models proposed in the literature can be seen in Table 2.

The literature presents various system designs aimed at addressing the challenge of phishing. However, many studies exhibit limitations by focusing solely on extracting features from isolated sections of emails such as the body, text, or header, thereby rendering these systems susceptible to exploitation despite achieving high accuracies on established datasets. In the dynamic threat landscape of contemporary phishing attacks, relying on a singular feature type within an email provides attackers with a single point of circumvention (Salloum et al., 2021).

To mitigate this vulnerability, hybrid feature extraction methodologies have been proposed, incorporating effective features from diverse sections of emails including headers, bodies, and URLs. This approach establishes multiple defencive layers, heightening the complexity required for an attack to evade detection. Hybrid models have demonstrated considerable success in the literature, achieving accuracies exceeding 99% on reliable datasets through various implementations and feature sets. Moreover, ensemble learning techniques have shown promise by leveraging different machine learning algorithms to optimise performance across varied feature sets (Bountakas and Xenakis, 2023). Studies employing ensemble methods indicate enhanced accuracy compared to single algorithm approaches (Anandita Yadav et al., 2017; Karim et al., 2023). Nevertheless, the limited presence of ensemble learning studies in phishing detection research necessitates further exploration into their efficacy when integrated with hybrid feature extraction, aiming to further elevate the accuracy of phishing detection systems. Hence, our paper proposes a novel ensemble learning phishing detection system using a stacking technique. This system employs a two-tiered stack, with two base learners chosen based on

Table 2
A comparative summary of related models from the literature.

Reference	ML model	Dataset	Number of features	Highest accuracy
Smadi et al. (2018)	Reinforcement learning, dynamically evolving NN	Phish: PhishingCorpus Legitimate: SpamAssassin URLS: Phishtank	50	98.63%
Hiransha et al. (2018)	CNN	IWSPA-AP Shared Task committee	–	96.80%
Alotaibi et al. (2020)	CNN	Phish: PhishingCorpus Legitimate: SpamAssassin	–	99.42%
Ramanathan and Wechsler (2013)	NLP + AdaBoost	Phish: PhishingCorpus Legitimate: SpamAssassin	–	96% F1-measure
Egozi and Verma (2018)	NLP + weighted DT	IWSPA-AP Shared Task committee	26	80% TPR
Sahingoz et al. (2019)	URL analysis with NLP + 7 ML algorithms comparison	Ebbu2017	–	97.98%
Gualberto et al. (2020)	LDA NLP + XGB	Phish: PhishingCorpus Legitimate: SpamAssassin	–	99.95% (validation)
Bountakas et al. (2021)	Word2Vec NLP + RF	Phishing: Nazario Legitimate: Enron	–	98.95%
Dewis and Viana (2022)	NLP + Deep Learning (LSTM/MLP)	Textual: SpamAssassin Numerical: Kaggle Email Spam	–	99% (textual) 94% (numerical)
Nguyen et al. (2014)	Novel URL analysis	PhishTank	6	97.16%
Smadi et al. (2015)	Data mining + 3 ML algorithms comparison	Phish: Nazario Legitimate: SpamAssassin	23	98.87%
Che et al. (2017)	Fuzzy	–	–	n/a
Nb and Kp (2018)	NLP + DT/RF	IWSPA-AP Shared Task committee	30	n/a (Overfitted model)
M. and Pais (2022)	FastText-CBOW + RF	Phish: PhishingCorpus Legitimate: HamCorpus	4	99.50% accuracy
Ma et al. (2009)	Hybrid feature extraction + DT	WestPac	7	99.80% (validation)
Rahmi et al. (2011)	Hybrid feature extraction + Bayesian Network	Phish: Nazario Legitimate: SpamAssassin	7	96%
Verma et al. (2012)	Hybrid feature extraction	Phish: Nazario	–	97.7%
Daeef et al. (2016)	Hybrid feature extraction + RF	Phish: Nazario Legitimate: SpamAssassin	48	99.91%
Niu et al. (2017)	Hybrid feature extraction + CS-SVM	Phish: Nazario + MillerSmiles Legitimate: Enron	23	99.52%
Yang et al. (2019)	Hybrid feature extraction + SVM	Phish: Nazario Legitimate: SpamAssassin	9	97.50%
Yadav and Panda (2021)	Hybrid feature selection + 3 ML algorithms comparison	Phish: Nazario Legitimate: SpamAssassin	15	99%
Ding et al. (2021)	Hybrid feature selection +KM-SMOTE + RF	Non-open source	27	97.16% F1-Measure
Anandita Yadav et al. (2017)	Ensemble: GNB, BNB, RF, KNN, SVM	SpamAssassin	15	98.02%
Kumar et al. (2020)	Probabilistic NN + SVM	Non-open source	–	98.3%
Karim et al. (2023)	Ensemble: LR, SVC, DT	Kaggle repository	33	98.12%
Bountakas and Xenakis (2023)	Ensemble: DT + KNN (soft-voting)	Phish: Nazario Legitimate: Enron + SpamAssassin	22	99.43%

their performance and distinct feature patterns. These base learners are combined using a meta-learner in the second tier for final classification. The model aims to improve accuracy by leveraging the strengths of multiple algorithms and incorporating hybrid feature selection for a comprehensive representation of the email.

3. Proposed ensemble learning model

3.1. Structure of the proposed model

The need for an effective phishing detection system arises from the limitations of traditional methods and the escalating sophistication of phishing attacks. While conventional ML approaches have improved accuracy, they often struggle to keep pace with evolving phishing tactics. Ensemble learning, by combining the strengths of multiple algorithms and utilising hybrid feature selection, offers a promising solution to enhance detection accuracy and mitigate the ever-growing threat posed by phishing emails (Bountakas and Xenakis, 2023). Stacking is an ensemble learning technique in which predictions from ML models called base learners are passed as input variables for models in a multi-layered stack (Pavlyshenko, 2018). A combination of different machine learning models can result in a significant increase in the accuracy of the system. This is especially true when the models predominantly rely on different subsets of the entire feature set, allowing one model to capture patterns in the data that are underutilised in the other model (Anandita Yadav et al., 2017).

We propose a novel stacking ensemble learning model for phishing detection to enhance the accuracy and robustness of existing solutions. A summarised diagram of the proposed model is shown in Fig. 1. The proposed model combines the strengths of multiple ML algorithms to achieve a more comprehensive and effective classification. The core of the stacking model lies in its hierarchical structure, where two base learners, RF and XGBoost, are trained on a rich set of features extracted from email content. These base learners provide diverse perspectives on the data, capturing different aspects of email characteristics. The predictions generated by the base learners are then fed into a meta-learner, Logistic Regression (LR), which acts as a final decision-maker. LR learns a linear relationship between the base learners' outputs and the true class label (phishing or legitimate), effectively combining their predictions to produce a more accurate final classification. This hierarchical structure allows the model to leverage the strengths of each algorithm, leading to improved performance. To further enhance the model's effectiveness, this paper emphasises the importance of hybrid feature selection. This approach involves extracting a comprehensive

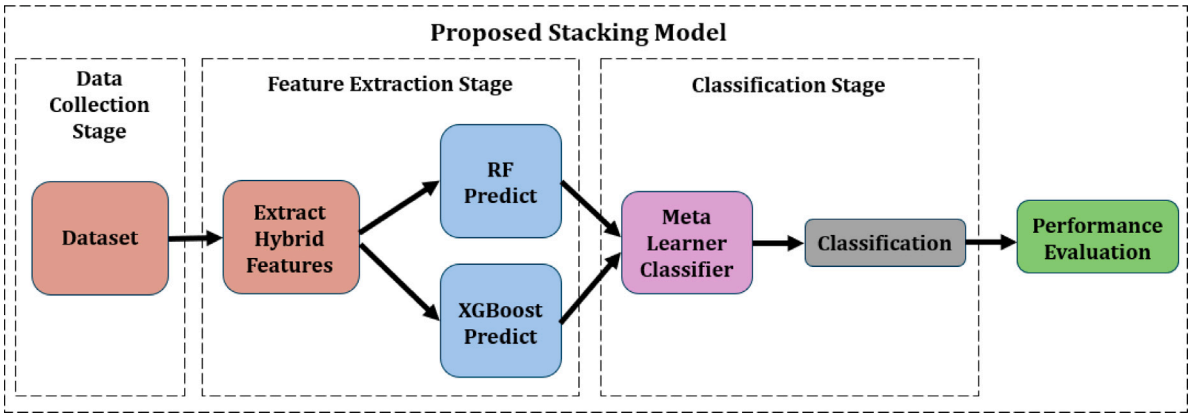


Fig. 1. Structural design of the proposed stacking model.

set of features from email content, encompassing lexical, syntactic, and URL features. Lexical features capture the vocabulary and word patterns within the email text, syntactic features analyse the grammatical structure, and URL features examine the characteristics of embedded URLs. By combining these diverse feature types, the model can identify phishing emails based on a holistic understanding of their content and structure. The structure of the proposed model can be shown in Fig. 2.

During the development phase, extensive testing was conducted to evaluate the impact of incorporating additional base learners to the ensemble. The results indicated that introducing models with lower individual performance adversely affected the overall efficacy of the proposed model. Furthermore, the inclusion of additional base learners significantly increased the processing time for each email, hindering the real-time applicability of the proposed model. This increase in computational demand would pose a challenge to the model's practical deployment, as it could hinder its ability to process emails in a timely manner. These findings highlight the crucial trade-off between model complexity and operational efficiency. This further highlights that careful consideration when selecting base learners with strong individual performance is crucial for maintaining the balance between classification accuracy and processing speed.

The flow of the proposed model started when input features were passed into two unique base learners, RF and XGBoost. These particular models are deployed in the proposed ensemble due to their high performance as individual models, alongside their unique properties of utilising different feature subsets, enabling the ensemble to effectively employ a wide range of the total feature set. Feature importance charts can be shown in Figs. 3 and 4. The disparity in the feature sets is most evident on the right side of the graph, where RF exhibits notably higher importance for these features. Furthermore, the total number of features employed by RF is greater than XGBoost.

Both base learners perform an initial classification of the email, predicting the probability that the email sample belongs to the positive class (phish). A threshold is then applied to the probabilities, resulting in binary predictions. The binary predictions are taken as input variables for the stacked meta learner which predicts the probability that the email sample belongs to the positive class. Finally, a threshold is applied to the probability outcome, resulting in the final classification. The implementation of the proposed stacking ensemble is presented as pseudocode in Algorithm 1.

To train the meta-learner, a suitable sample of accurate predictions from both the RF and XGB models is needed. Hence, the test predictions of the individual base learners are sampled with a train/test split. This provides the meta-learner with accurate predictions from previously trained base learners. Fig. 5 shows the training of the proposed model.

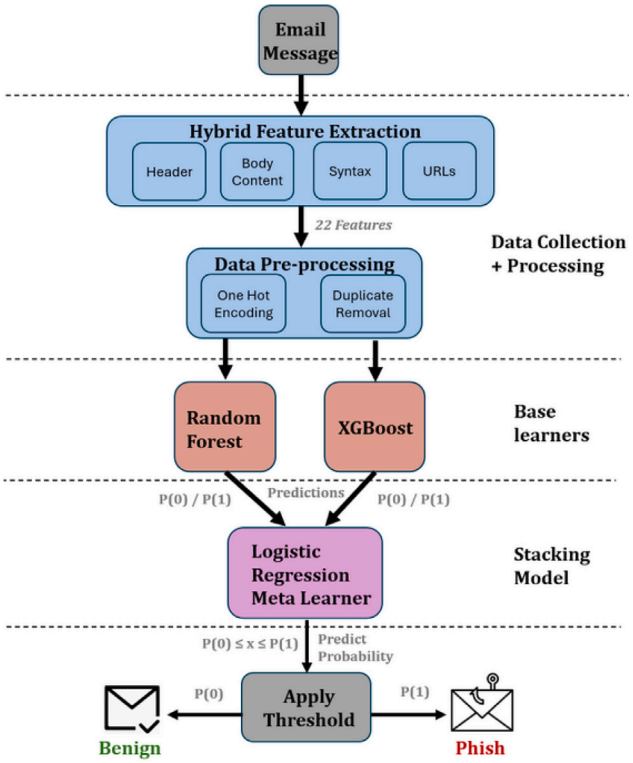


Fig. 2. Flow of the proposed ensemble learning stacking model.

3.2. Feature extraction

Feature extraction is crucial in phishing detection as it transforms raw email data into meaningful attributes that ML models can effectively analyse. This process involves identifying and extracting relevant features from emails, such as text content, URL characteristics, sender information, and email header details. By effectively selecting and extracting these features, the model gains a comprehensive understanding of the email's characteristics, enabling it to distinguish between legitimate and phishing emails. Effective feature extraction is critical for improving the accuracy and robustness of phishing detection systems, ultimately protecting users and organisations from malicious attacks (Pavlyshenko, 2018). The feature extraction technique deployed in this study is designed to operate directly on raw email messages.

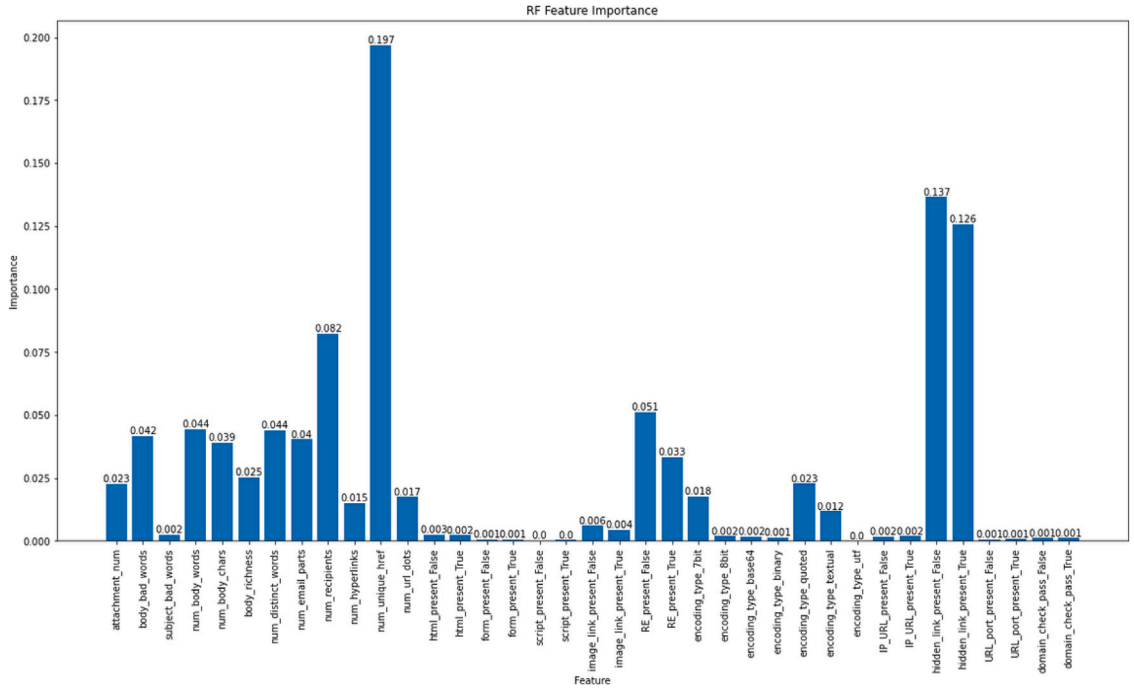


Fig. 3. Graphical representation of feature relevance based on Random Forest feature importance values.

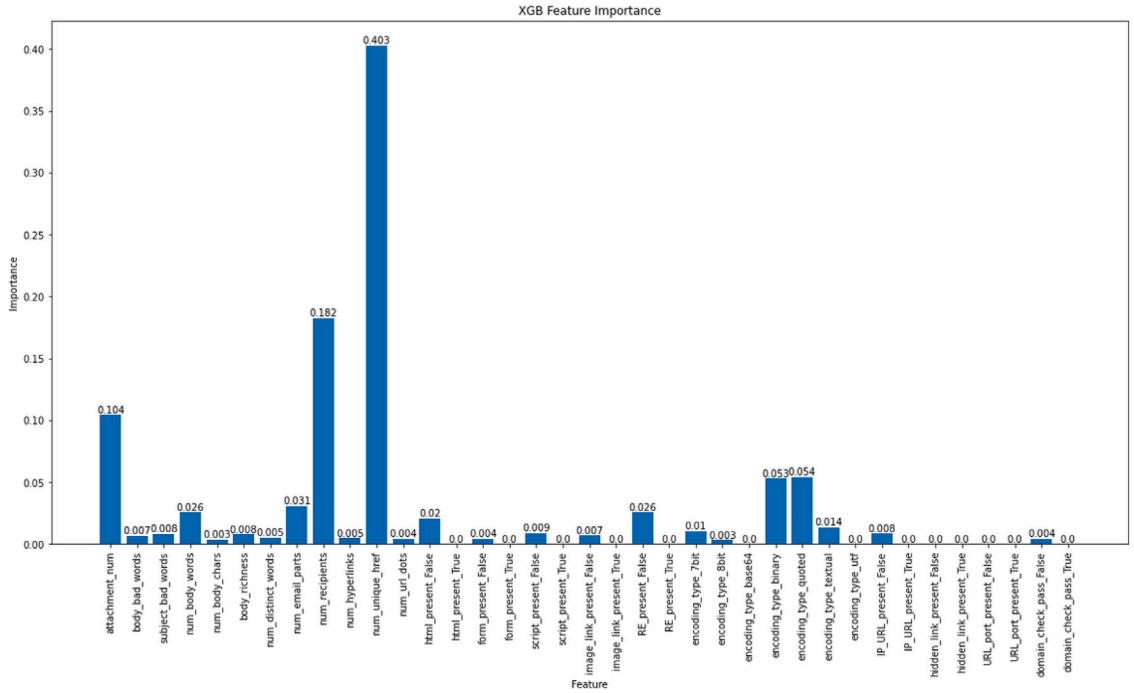


Fig. 4. Graphical representation of feature relevance based on XGBoost feature importance values.

This methodology enhances the applicability of the model across various email platforms, eliminating dependence on platform-specific features and ensuring suitability for deployment in diverse real-world environments.

Our paper proposed a hybrid feature extraction technique in which a selection of high-quality features was considered from various email sections such as body content, body syntax, header and URLs, culminating in a total of 22 features. Hybrid feature extraction combines the strengths of multiple sets of features captured across multiple email sections. By leveraging various feature types, the resulting model can capture diverse patterns overlooked in alternative feature extraction techniques of which the resulting model is trained on only a single feature type. This technique improves the robustness and generalisability of the model, resulting in higher performance, particularly in complex datasets where valuable information is distributed in different email sections (Khonji et al., 2013). Furthermore, NLP techniques utilised frequently in the literature fail to capture the intrinsic patterns present in the email headers and URLs due to their heavy reliance on textual data, where phishing emails often contain limited, generic or misleadingly benign text (Bountakas et al., 2021).

Algorithm 1 Proposed Stacking Ensemble

Input: Training dataset D_{train}, test dataset D_{test}
Output: Meta leaner classifications $y_{\text{classification}}$
Train Random Forest model RF using training dataset D_{train}
for each tree T in forest F **do**
    $D^{(i)} \leftarrow$ Bootstrap sample from D
    Train decision tree $T_i \leftarrow D^{(i)}$ with m randomly selected features
    $RF\{T_1, T_2, ..., T_N\} \leftarrow$ Add T_i
end for
Train XGBoost model XGB using training dataset D_{train}
for each tree T in ensemble E **do**
    Compute gradient G and hessian H for each instance in D_{train}
    Train $T \leftarrow G$ and H as input features
    $XGB\{T_1, T_2, ..., T_N\} \leftarrow$ Add T_i
end for

for each instance x in D_{test} **do**
    Predict class probabilities P_{RF} with RF for x
    for T_i in $RF\{T_1, T_2, ..., T_N\}$ **do**
        $P_i \leftarrow$ Predict class probabilities using tree T_i
        $RF\{P_1, P_2, ..., P_n\} \leftarrow P_i$
    end for
    $P_{\text{RF}} \leftarrow \text{AVG}(RF\{P_1, P_2, ..., P_n\})$

    Predict class probabilities P_{XGB} with XGB for x
    for T_i in $XGB\{T_1, T_2, ..., T_N\}$ **do**
        $P_i \leftarrow$ Predict class probabilities using tree T_i
        $XGB\{P_1, P_2, ..., P_n\} \leftarrow P_i$
    end for
    $P_{\text{XGB}} \leftarrow \text{AVG}(XGB\{P_1, P_2, ..., P_n\})$

    Combine P_{RF} and P_{XGB} into feature vector F
end for

Combine feature vector F and labels in D to create meta dataset M
Split M into M_{train} and M_{test}

Train logistic regression meta learner LR using M_{train}
for each instance x in D_{train} **do**
    Predict class probability $P \leftarrow$ Evaluate gradient of cost function
    Apply learning rate
end for

for each x in M_{test} **do**
    Predict class probability P using LR for x
    Append P to $y_{\text{prediction}}$
    Apply Threshold to $y_{\text{prediction}} \rightarrow y_{\text{classification}}$
end for
return $y_{\text{classification}}$

The features used are well-established and have been applied in various hybrid phishing detection systems with notable accuracy such as the work proposed by Bountakas and Xenakis (2023) in which these features are brought together. This highlights their reliability and effectiveness in capturing different aspects of email messages. The implementation of the feature extraction can be shown in Fig. 6, displaying an illustrated representation of the process. Pseudocode is additionally displayed in Algorithm 2, highlighting a simplified presentation of the feature extraction code. The email set is split into individual emails, each email is parsed in turn, extracting features across header, body content, syntax and URL categories. For each email, the extracted features are stored in a list and appended to a CSV file

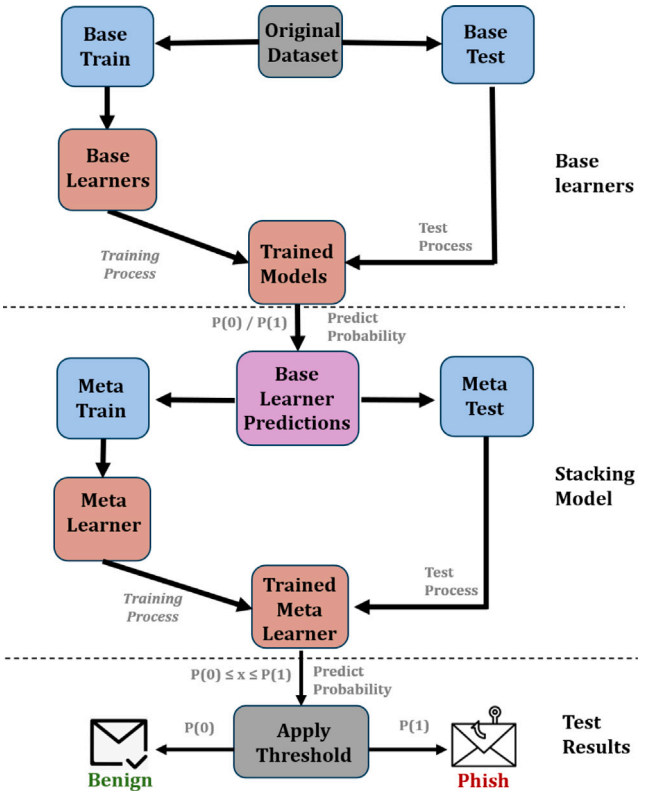


Fig. 5. The training and testing methodology adopted for the proposed stacking model.

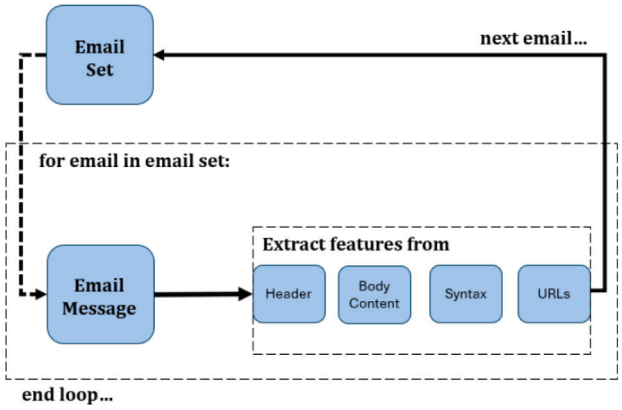


Fig. 6. Flow of the feature extraction technique of the proposed stacking model.

before iterating to the next email. A brief summary of the selected features can be shown in Table 3.

Algorithm 2 Feature Extraction

Input: Raw email Set E
Output: CSV of extracted features F
for each email e in email set E **do**
    Extract $header$ features as h
    Extract $body$ features as b
    Extract $syntax$ features as s
    Extract url features as u
    List L of features $\{F_1, F_2, ..., F_N\} \leftarrow h, b, s, u$
    Write L to CSV file F
end for

The feature extraction process involves a multi-faceted approach, encompassing various aspects of an email to provide a comprehensive representation. Firstly, features are extracted from the email body, as a variety of different techniques are employed by threat actors to increase the chance of luring a victim to provide sensitive information or download malware. The email body is analysed for the presence of common phishing techniques, including urgency, threats, and promises of rewards, all designed to manipulate the recipient into taking action. To maximise the detection performance, a careful selection of these features is conducted, ensuring they effectively capture the subtle nuances of phishing emails. For instance, binary features indicating the presence or absence of phishing tactics are combined with numerical features, such as the frequency of specific keywords or the length of suspicious URLs embedded in the content. These features are selected based on their demonstrated impact on previous phishing detection systems and their ability to generalise across diverse datasets.

The inclusion of these features plays a critical role in enhancing the model's predictive accuracy, as each feature contributes unique insights into different facets of phishing behaviour. For example, while urgency and threats are indicative of social engineering tactics, features like the structure of embedded URLs and syntactic anomalies in the email text help identify technical inconsistencies often overlooked by recipients. By combining these complementary features, the feature extraction process achieves a holistic representation of phishing characteristics, significantly improving the model's detection capabilities (Ma et al., 2009; Rahmi et al., 2011; Bountakas and Xenakis, 2023). This category of features checks for the presence of several of these techniques, with a true or false binary extraction. Where necessary an integer value is utilised to count the number of occurrences in the email.

1. **Html_present**: This feature extracts the presence of HTML code in an email.
2. **Form_present**: This feature records the presence of HTML forms in an email, which are often implemented in phishing campaigns to extract sensitive information.
3. **Script_present**: This feature highlights the presence of JavaScript (JS) code in the email. JS is commonly deployed in phishing emails, and is identified through the use of the HTML script tag.
4. **Attachment_num**: This feature represents an integer count of the number of email attachments, commonly used to distribute malware in phishing campaigns.
5. **Image_link_present**: This feature highlights the presence of a 'hidden' hyperlink added to an image. Malicious hyperlinks are often hidden behind seemingly benign images to increase the likelihood of a user clicking the link.

Secondly, the syntactic traits of the email are examined to identify linguistic patterns commonly found in phishing emails. This approach leverages the understanding that phishing emails, despite mimicking legitimate emails, often exhibit subtle differences in language usage due to their deceptive nature. These features capture critical discrepancies in the structure and formatting of phishing emails, which often exhibit overly verbose or unusually concise language. Enabling simple features such as 9 and 10 which are integer counts, which might appear trivial initially, to identify critical structural anomalies commonly found in phishing emails.

6. **Body_bad_words**: This feature represents an integer count of the following words in the email body text, "link", "click", "confirm", "user", "customer", "client", "suspend", "restrict", "verify", "payment", "protect". These words have been identified to appear frequently in phishing emails in various related studies (Ma et al., 2009; Rahmi et al., 2011; Bountakas and Xenakis, 2023).
7. **Subject_bad_words**: Similar to the previous feature, however, the occurrences of these words in the subject are counted, as opposed to the body text.
8. **RE_present**: Identifies the presence of "RE" at the beginning of the email subject. RE in this context, is used to denote that an email is replying to a previous email. Phishing emails are usually not replying to previous emails and therefore commonly do not contain this abbreviation.
9. **Num_body_words**: An integer count of the total number of words in the email body.
10. **Num_body_chars**: An integer count of the total number of characters in the email body.
11. **Body_richness**: Using the previous two features, the ratio of words to characters is calculated.

$$\text{Richness} = \frac{\text{No. of words in email body}}{\text{No. of characters in email body}}$$

12. **Num_distinct_words**: This features provides an integer count of the number of distinct words in the email body.
13. **Num_email_parts**: This feature provides an integer count of email sections, e.g text, html.

Thirdly, the email header is analysed for specific features that can reveal subtle patterns missed by analysing the email body alone. While limited to two features, this step helps in identifying potential red flags.

14. **Encoding_type**: This feature extracts the content-transfer-encoding-header field, used to specify the auxiliary encoding type applied to the email data, which enables it to pass through mail transport mechanisms that may contain limitations on character set (Borenstein and Freed, 1992).
15. **Num_recipients**: An integer count of the number of recipients the email was sent to. It is common for phishing campaigns to target multiple/many users at the same time.

Finally, the URLs embedded within the email are analysed for discrepancies in their usage compared to legitimate emails. This category of features aims to detect malicious URLs, which are a common tactic employed in phishing attacks.

16. **IP_URL_present**: This feature tracks the presence of a URL that utilises an IP address instead of a domain name. It is uncommon for legitimate URLs to contain IP addresses. However, temporary malicious URLs will often utilise an IP address.
17. **Num_hyperlinks**: This feature will count the number of hyperlinks present in the email.
18. **Hidden_link_present**: This feature identifies the use of clickable text to hide a hyperlink.
19. **Num_unique_href**: This feature extracts an integer count of the number of different URLs in the email.
20. **Num_url_dots**: This feature tracks the total number of dots found in URLs throughout the email body.
21. **URL_port_present**: Similarly to feature 16, this feature checks whether URLs in the email contain a port number, as this is uncommon in legitimate URLs.
22. **Domain_check**: This feature represents a boolean check, whether the domain of the email sender matches the domains deployed in URLs in the email.

By combining these four categories of features, the proposed ensemble learning model achieves a comprehensive representation of an email, enabling it to more effectively identify phishing attempts. This hybrid feature selection approach contributes significantly to the models high accuracy in detecting phishing emails (Rahmi et al., 2011).

Table 3
Summarised description of hybrid features extracted from raw email samples.

ID	ML Model	Description	Data type
1	html_present	Presence of HTML in the email body	Boolean
2	form_present	Presence of HTML form in email body	Boolean
3	script_present	Presence of HTML script in email body	Boolean
4	attachment_num	Number of email attachments	Integer
5	image_link_present	Presence of an image containing hyperlink	Boolean
6	body_bad_words	Count of 'bad words' in email body	Integer
7	subject_bad_words	Count of 'bad words' in email subject line	Integer
8	RE_present	Presence of 'RE' in the emails subject line	Boolean
9	num_body_words	Count of words in the emails body	Integer
10	num_body_chars	Count of characters in the emails body	Integer
11	body_richness	Ratio of words to characters in emails body	Float
12	num_distinct_words	Count of distinct words in the emails body	Integer
13	num_email_parts	Number of different email sections	Integer
14	encoding_type	Encoding type of the email	String
15	num_recipients	Total number of recipients in the email	Integer
16	IP_URL_present	Presence of an IP address in a URL	Boolean
17	num_hyperlinks	Count of hyperlinks in the email	Integer
18	hidden_link_present	Presence of hyperlink hid behind text	Boolean
19	num_unique_href	Count of the unique links in the email	Integer
20	num_url_dots	Total count of dots '.' in URLs	Integer
21	URL_port_present	Presence of a port number in a URL	Boolean
22	domain_check	Does the domain name in the sender address match the domains in any URLs	Boolean

3.2.1. Data preprocessing techniques

Data preprocessing techniques play a crucial role in enhancing the effectiveness of feature extraction for phishing detection. These techniques bridge the gap between raw email data and the structured information required by machine learning models (Ramanathan and Wechsler, 2013; Salloum et al., 2021). Firstly, data preprocessing ensures consistency and uniformity in the extracted features. Raw email data often exhibits inconsistencies in formatting, capitalisation, and punctuation. Techniques like text normalisation, stemming, and lemmatisation address these issues by transforming words into their canonical forms, removing irrelevant variations, and standardising the representation of words (Alotaibi et al., 2020). This standardisation process facilitates accurate feature extraction and improves the model's ability to identify patterns across different emails (Rahmi et al., 2011). Hence, duplicate rows are removed to ensure emails originating from the same phishing campaign are not included multiple times in the data, providing an accurate representation of the model's inherent ability to generalise to new unseen data as the possibility of providing the model with a sample it has seen previously has been removed (Gudivada et al., 2017).

Secondly, data preprocessing plays a critical role in addressing potential biases inherent in the dataset (Bountakas and Xenakis, 2023). Phishing emails typically constitute a smaller proportion of the overall email population, leading to class imbalance. This imbalance can skew the training process, causing the model to favour the majority class (legitimate emails) and underperform on the minority class (phishing emails). To mitigate this issue, an initial preprocessing step involved randomly shuffling the dataset to ensure that the email samples were not arranged in chronological order. This measure aimed to minimise the risk of temporal bias, which could arise if older samples were predominantly used for training and newer samples for testing (Ramanathan and Wechsler, 2013). Without this step, the model might inadvertently learn temporal dependencies, such as associating certain features with older emails, which could reduce its ability to generalise effectively to recent phishing tactics. By ensuring that training and testing data are randomly distributed, shuffling helps the model focus on identifying patterns intrinsic to phishing emails rather than relying on outdated or irrelevant temporal cues, ultimately enhancing its robustness and adaptability to evolving threats.

Finally, data preprocessing enhances the efficiency of feature extraction, so one-hot encoding was employed for non-numeric columns ensuring that the data is in the correct format. One-hot encoding was further applied to the boolean columns, preventing the introduction of potential bias through the inclusion of unwanted ordinal relationships in the data (Piepenbreier, 2023). The incorporation of the one-hot encoded columns comes with an expanded data dimensionality with a total of 37 columns. The overall implementation of the pre-processing steps was summarised in Fig. 7.

Table 4
Experiment settings highlighting the development environment utilised to develop, train and test the proposed stacking model.

Development Virtual Machine	
Operating System	Ubuntu 22.04 LTS (Linux x86_64)
Base Memory	8192 MB
Processors	6
Host Machine	
Operating System	Windows 10
Memory	15900 (Usable)
Processors	16
Graphics Card	NVIDIA GeForce RTX 2070
Virtualisation Software	Virtualbox 7.0.8

4. Experimental results

This section presents experiment settings, evaluation metrics, datasets and the results achieved by the proposed stacking ensemble learning technique.

4.1. Experiment settings

The development environment to implement this proposed stacking phishing detection system was an Ubuntu 22.04 LTS virtual machine. The specifications of both the virtual machine and the host machine are detailed in Table 4. The system was developed using Python 3, leveraging a suite of libraries for various tasks. These libraries, along with their functionalities, are listed in Table 5.

4.2. Evaluation metrics

The performance of the phishing detection system was evaluated using a set of standard metrics: accuracy, precision, recall, and F1-score. These metrics are widely adopted in the phishing detection literature, with accuracy typically considered the most significant measure (Varoquaux and Colliot, 2023; Khonji et al., 2013; Salloum et al., 2021). The formulas used to calculate these metrics are presented below, where the True Positives (TP) are the number of correctly classified

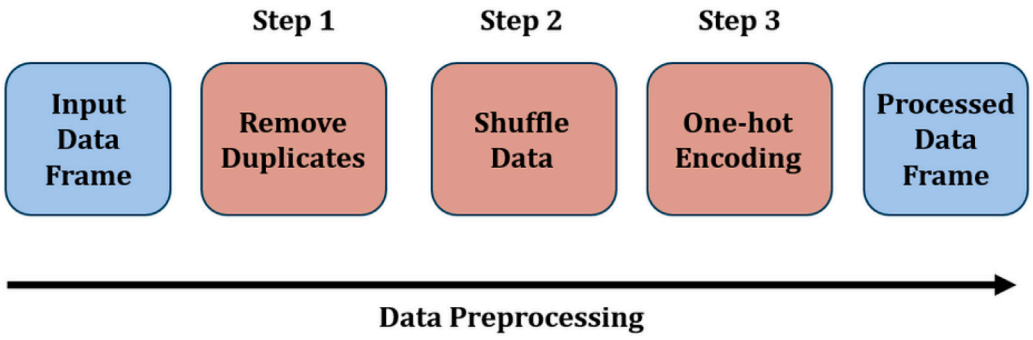


Fig. 7. High level overview of the data preprocessing strategy adopted in the proposed stacking model.

Table 5
Python libraries used in the implementation of the proposed stacking model.

Import	Description
Jupyter (Jupyter, 2015)	Combining markdown and code in a notebook
Beautiful Soup 4 (Richardson, 2015)	Library for parsing and extracting data from HTML
Urllib (Python)	Module in urllib library used to parse URLs
Pandas 1.4.2 (Pandas, 2022)	Library used for data analysis
Scikit-learn (Scikit-learn)	Library for machine learning and data analysis
NumPy (Numpy, 2023)	Library used for numerical operations
MatplotLib (Matplotlib)	Library used for creating figures
Seaborn (seaborn)	Library for creating graphics
XGBoost 2.0.3 (XGBoost, 2022)	Package utilised to implement XGBoost algorithm

phishing emails, False Positives (FP) are the number of misclassified benign emails, True Negatives (TN) are the number of correctly classified benign emails, and the False Negatives (FN) are the number of misclassified phishing emails (Bountakas et al., 2021).

$$(1)\ \textbf{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$(2)\ \textbf{Precision} = \frac{TP}{TP + FP}$$

$$(3)\ \textbf{Recall} = \frac{TP}{TP + FN}$$

$$(4)\ \textbf{F1 - Measure} = 2 * \frac{precision * recall}{precision + recall}$$

In addition to the primary performance metrics, this paper also considered the Receiver Operating Characteristic (ROC) curve and detection time. While not as extensive in the literature, the ROC curve provides valuable insights by plotting the sensitivity (True Positive Rate, TPR) against the specificity (True Negative Rate, TNR) at various threshold values (Hoo et al., 2017). Although it is not commonly discussed in the literature, detection time is a crucial factor when deploying ensemble models. The use of multiple ML models inherently leads to increased processing time. Measuring detection time allows for monitoring this effect and ensuring the system's practicality in real-world scenarios. Furthermore, a confusion matrix was generated for each model, providing a visual representation of their performance (Anandita Yadav et al., 2017). This visualisation aids in understanding the model's ability to correctly classify phishing and legitimate emails, offering a comprehensive view of its strengths and weaknesses (Rahmi et al., 2011).

4.3. Datasets

The datasets selected for legitimate emails were the Apache SpamAssassin Public Corpus (Apache, 2004) and Enron public email dataset (CMU, 2011). Phishing emails were sourced from the Jose Nazario phishing dataset (Nazario, 2024). These datasets were selected because they are widely recognised and frequently cited in the literature, facilitating a robust comparison between the performance of the proposed stacking model and existing state-of-the-art phishing detection systems. Their established usage ensures consistency and credibility when benchmarking results. However, we acknowledge that these datasets have inherent limitations that affect the evaluation of phishing detection models. While valuable, they do not fully represent the dynamic and rapidly evolving phishing landscape. The relatively small sample size and the absence of zero-day phishing emails limit their ability to capture the diversity and sophistication of real-world threats (Yadav and Panda, 2021). Additionally, the widely used and well-established public datasets are further limited by the absence of attachment samples, a key feature for enhancing phishing detection and detecting real-world phishing threats (Dewis and Viana, 2022). These shortcomings reflect a broader issue in the phishing detection domain: the lack of comprehensive, up-to-date datasets that include a wide range of contemporary and emerging phishing techniques. Such limitations hinder the development and testing of models capable of generalising effectively to real-world scenarios, particularly in live deployment environments. Addressing this challenge will require a concerted effort to develop and share richer, more representative datasets that mirror the current threat landscape. This would significantly enhance the applicability and reliability of research-driven phishing detection systems.

In the final dataset, the total number of email samples was 12094. The split between phishing and benign emails is approximately balanced with 5874 legitimate emails and 6220 phishing emails, a ratio of 2.9:3.1. The final dataset deployed follows the trend in the literature, deploying open source datasets used in similar systems developed by Rahmi et al. (2011), Verma et al. (2012), Smadi et al. (2015), Daeef et al. (2016), Anandita Yadav et al. (2017), Yang et al. (2019), Yadav and Panda (2021), Bountakas et al. (2021) and Bountakas and Xenakis (2023). The literature review revealed a prevailing trend where the majority of studies in the literature utilise balanced datasets. In alignment with this trend, the final dataset deployed in this study was balanced, allowing for meaningful comparisons with existing research (Bountakas and Xenakis, 2023). The train/test split deployed on the full dataset was 65% train and 35% test (Alhogail and Alsabih, 2021). Various split ratios were tested, and this split achieved the highest overall performance across the different models. Additionally, a large test set enabled a significant improvement in the accuracy of the stacking model, as this provides the meta learner with a larger number of data samples for both training and testing.

The base learners are trained on the full dataset, which comprises 12,094 emails, enabling them to capture a comprehensive range of

patterns and features for phishing detection. The stacking model, as a higher-level ensemble, relies on the precise predictions generated by its base learners during both the training and testing phases. To achieve this, a subset of the original test data used by the base learners is designated for training and testing the meta learner. Specifically, the meta learner operates on a dataset of 4,232 emails, with 2,962 emails allocated for training and 1,270 for testing. The proposed model utilises the same dataset as the individual base learners. However, it is a subsample of the original testing set. This approach is inherent to the nature of the stacking ensemble methodology. This design ensures that the meta learner utilises the predictions of the base learners to refine its decision-making process, leveraging their strengths while compensating for their individual weaknesses. By training on a subsample derived from the original test set, the stacking ensemble adheres to the methodology's core principle of combining diverse base-level outputs to achieve superior predictive performance. This structured approach allows the meta learner to focus on learning from the relationships and interactions among the base learners' predictions, optimising its ability to generalise effectively to unseen data. This requires clear separation between the data used to train the base learners and the data used to train and evaluate the meta learner, ensuring that the meta learner is trained with predictions from the base learners that were not trained on the same email samples.

4.4. Results

The proposed stacking ensemble model achieved exceptional performance, demonstrating its effectiveness in detecting phishing emails. The model achieved the highest overall performance when compared against the individual ML algorithms, gaining an accuracy of 99.53%. This high accuracy indicates the model's ability to correctly classify a vast majority of emails as either legitimate or phishing. The accuracy shows the proposed model is highly reliable, and would be a valuable tool offering a robust solution for real-world email security applications in which classification accuracy is critical for both preventing phishing attacks and ensuring minimal disruption to legitimate email flow. Additionally, the model attained a precision of 99.25%, recall of 99.85% and an F1-Measure of 0.9955. The increase in recall demonstrates the model's enhanced ability to identify phishing emails, which is crucial for minimising false negatives and ensuring that phishing emails are not missed resulting in a security incident. The cost of missing a phishing email can be substantial, and the model's ability to capture a higher proportion of phishing emails reinforces its suitability for real-world applications, where detecting as many phishing attempts as possible is of utmost importance. The model's high precision value displays that it rarely misclassifies legitimate emails as phishing. This reduces the risk of false positives, which can lead to unwanted inconvenience and frustration through the disruption of legitimate email flow by blocking benign emails. The proposed model strikes an important balance, ensuring both a high precision and recall, leaning slightly towards recall by capturing most phishing attempts while maintaining minimal impact on benign communications.

The significant gains in both recall and precision achieved by the proposed model highlight its ability to effectively identify and block a substantial number of phishing emails when scaled to larger datasets or real-world deployments. These complementary strengths make the combination of RF and XGB highly effective, with both models leveraging their unique traits to enhance the overall performance of the system. Integrating multiple models within the stacking ensemble enables the proposed model to capitalise on the individual strengths of each algorithm while minimising their weaknesses. This synergy results in a reliable and effective phishing detection mechanism capable of reaching high accuracies even when compared to the state-of-the-art phishing detection systems from the literature.

This makes the model highly practical for deployment in the real-world, where the consequences of missing a phishing email are significantly more damaging than mistakenly blocking a legitimate email.

	precision	recall	f1-score	support
0	0.9983	0.9917	0.9950	605
1	0.9925	0.9985	0.9955	665
accuracy			0.9953	1270
macro avg	0.9954	0.9951	0.9953	1270
weighted avg	0.9953	0.9953	0.9953	1270

Fig. 8. Raw results produced by the proposed stacking model output.

In a phishing detection system, the primary concern is to minimise the risk of phishing emails arriving in end user mailboxes due to the risk of security breaches. While blocking legitimate emails can cause inconvenience, it is a much less critical issue compared to the potential harm caused by undetected phishing attacks.

The F1-measure of 0.9955 further emphasises the model's effectiveness. This metric combines precision and recall, providing a balanced measure of the model's overall performance. The high F1-measure indicates that the model effectively captures positive cases (phishing emails) with a low number of misclassified positive samples. This balance between precision and recall is crucial for a robust phishing detection system, ensuring both high accuracy and minimal false alarms. The raw results produced by the proposed stacking model can be shown in Fig. 8. These results provide compelling evidence of the stacking ensemble's effectiveness, highlighting its reliability and robustness in predictions. The model's ability to achieve such high performance on unseen, real-world email samples demonstrates its potential for practical application in combating the ever-evolving phishing threat.

The proposed model's performance on the test set, comprising 1270 emails, demonstrates its effectiveness in identifying phishing emails. As illustrated in Fig. 9, the confusion matrix highlights the model's exceptional recall, with only one phishing email being misclassified. This remarkable recall signifies the model's proficiency in capturing phishing emails, prioritising their detection over potentially misclassifying legitimate emails. This characteristic is particularly valuable in a phishing detection system, as a single undetected phishing email can inflict significant harm on individuals or organisations.

Furthermore, the model's impressive precision is complemented by its high recall, with only 5 legitimate emails being misclassified as phishing. This balance between precision and recall underscores the model's ability to minimise both false positives and false negatives. The model's capacity to limit misclassification to a single phishing email emphasises its significant contribution to mitigating the phishing problem. The model's performance is further substantiated by its high Area Under Curve (AUC) value of 0.997, as shown in Fig. 10. This metric indicates the model's ability to effectively differentiate between positive (phishing) and negative (legitimate) data samples across a range of threshold values. The high AUC value highlights the model's strong discriminative power, enabling it to reliably separate phishing emails from legitimate emails.

The performance of the individual models including RF, DT, XGBoost, AdaBoost, LR and KNN is displayed in Table 6 alongside the proposed stacking ensemble learning technique. The results are further presented graphically in Fig. 11. Among the individual models, RF and XGBoost emerged as the top performers, both achieving an accuracy of 99.10%. While both models exhibited high accuracy, the RF model demonstrated slightly better performance in classifying phishing samples, as evidenced by its higher recall. Conversely, the XGBoost model exhibited a more balanced performance, with a lower number of misclassified legitimate emails.

While DT achieved the fastest detection time at an impressive 0.23 ms, its accuracy fell short of the performance achieved by RF and XGBoost. Although such a rapid detection time is commendable,

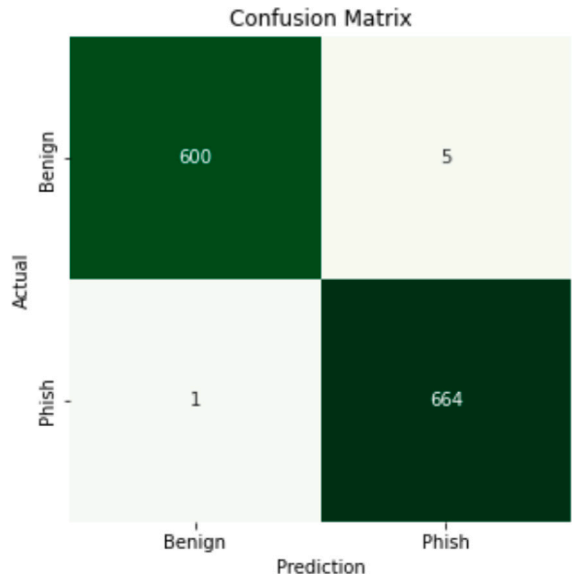


Fig. 9. Confusion matrix of the proposed stacking model.

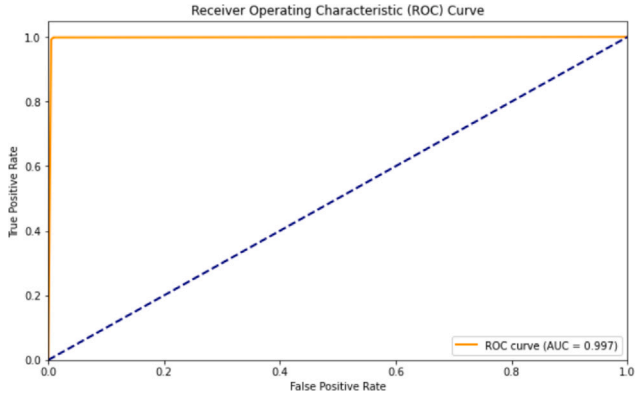


Fig. 10. Graphical representation of the ROC curve, illustrating the classification effectiveness of the proposed stacking model.

the classification performance of the DT model is insufficient to offset this advantage. In a security system, where accuracy is paramount, speed alone cannot justify the adoption of a model that fails to deliver reliable detection results. The DT model, despite its time efficiency, does not exhibit the complexity and granularity required to effectively identify the intrinsic patterns that distinguish phishing emails from legitimate emails with as much accuracy as more complex models such as RF and XGB. Its simplicity, while it contributes to faster processing times, ultimately limits DT's suitability for real-world phishing detection application, where even a minor sacrifice in accuracy can have severe consequences. This disparity in performance is attributed to the inherent strengths of the RF and XGB algorithms, which are known for their robust classification capabilities. Due to the impressive performance of the RF and XGB algorithms, they were deployed as base learners in the stacking ensemble model as utilising the most effective algorithms for the ensemble will increase the performance of the resulting ensemble system.

LR and KNN, known for their simplicity, exhibited the least effective performance among the individual classifiers. These models, due to their inherent simplicity, struggled to capture the intricate complexity of the feature set, leading to subpar performance. This observation highlights the significance of selecting ML models that are well-suited to the complexity of the data. The results highlight the importance of deploying models capable of delving deeper into the feature set to extract intricate patterns, such as Random Forest (RF) and boosting techniques. Employing weaker classifiers as base learners in an ensemble system would likely result in lower classification effectiveness compared to utilising stronger, more complex algorithms allowing the system to thoroughly analyse the feature set, identifying and extracting patterns with a higher reliability. By leveraging advanced algorithms, the stacking system can discern subtle and complex relationships within the data, enabling it to more effectively differentiate between phishing and legitimate emails.

The proposed stacking ensemble learning approach demonstrated superior performance across all detection metrics, consistently outperforming individual ML algorithms. Particularly, it achieved an accuracy of 99.53%, surpassing the highest-performing individual ML algorithm by a significant margin of 0.43%. This substantial improvement in accuracy signifies a notable enhancement in the model's ability to accurately detect and classify phishing emails, as a result strengthening its capacity to identify and mitigate potential security threats. Furthermore, this performance gain was achieved with a minimal increase in detection time of only 1.6 ms. These findings highlight the potential benefits of employing ensemble learning techniques in addressing the phishing problem, as they demonstrate that significant performance enhancements can be realised without compromising on computational efficiency.

4.5. Evaluation of proposed model

The proposed stacking ensemble learning model for phishing email detection demonstrates significant improvements over existing work in the literature. While various ML techniques have been employed for phishing detection, the model's hybrid feature selection and two-tiered stacking approach contribute to its exceptional performance. The performance of the proposed model was evaluated alongside recent comparable work from the literature as shown in Table 7.

The proposed stacking ensemble learning model demonstrates a significant leap forward in phishing detection accuracy, outperforming even the most effective systems documented in recent literature. This remarkable achievement is attributed to several key factors. Firstly, the model achieved an impressive accuracy of 99.53%, surpassing the highest accuracy of 99.10% reported for any individual ML algorithm. This substantial improvement highlights the potential of ensemble learning in tackling the ever-evolving challenge of phishing detection.

The proposed model achieves an average processing time of 3.7075 ms per email, which is sufficiently fast for real-world application, particularly in environments where rapid detection is essential. As we have introduced detection time as a novel performance metric, an aspect not previously emphasised in past phishing detection literature. Therefore, it is not feasible to compare the processing speed of the proposed model against existing models in the literature. At its current processing speed, the proposed model is capable of classifying approximately 270 emails per second, demonstrating its potential for high-throughput systems. This scalability feature is a crucial advantage, particularly for organisations handling vast volumes of email traffic daily. The low detection latency ensures that the model can provide timely responses, reducing the window of opportunity for phishing attacks to succeed. The model's scalability is further supported by its lightweight computational design, which allows it to maintain its performance across varying hardware configurations. This adaptability ensures that the proposed model can be deployed effectively in diverse environments, ranging from small-scale systems with limited resources to large-scale cloud-based infrastructures. The proposed stacking models efficiency is enabled through the use of LR, a fast and lightweight algorithm as the meta learner, cutting detection time while upholding a high accuracy. Although DT and KNN achieve even faster detection speeds due to their simplicity, they lack the detail and ability to capture complex patterns

Table 6
Comparison of experimental results between the proposed ensemble learning model and individual single layered classifiers.

Model	Accuracy	Precision	Recall	F1-measure	Detection time (ms)	AUC	TP	TN	FP	FN
RF	0.9910	0.9899	0.9927	0.9913	2.1013	0.999	2148	2047	22	16
DT	0.9802	0.9819	0.9792	0.9806	0.2274	0.982	2119	2030	39	45
XGB	0.9910	0.9917	0.9908	0.9912	1.5658	0.999	2144	2051	18	20
ADA	0.9889	0.9885	0.9898	0.9891	7.6601	0.998	2142	2044	25	22
LR	0.9577	0.9653	0.9515	0.9583	0.9866	0.990	2059	1995	74	105
KNN	0.9535	0.9703	0.9376	0.9537	0.5811	0.971	2029	2007	62	135
Proposed Model	**0.9953**	**0.9925**	**0.9985**	**0.9955**	3.7075	0.997	664	600	5	1

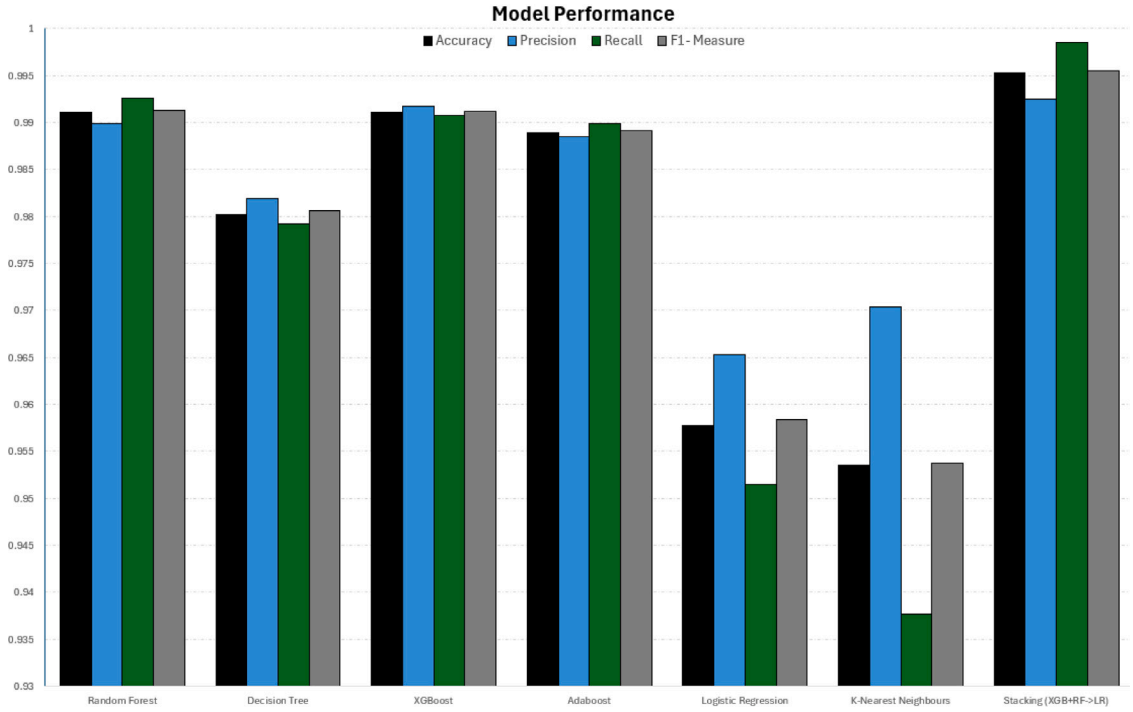


Fig. 11. Performance comparison between proposed ensemble learning model and individual classifiers.

Table 7
Evaluation of the proposed ensemble learning model and related studies in the literature.

Model	Model design	Accuracy	F1-measure	Detection time (ms)	Benign samples	Phish samples
Yadav and Panda (2021)	Hybrid	99.00%	Not provided	Not provided	500	500
Alotaibi et al. (2020)	Deep Learning	99.42%	0.9917	Not provided	6047	7315
Bountakas et al. (2021)	NLP	98.95%	0.9897	Not provided	14 000	1407
Karim et al. (2023)	Ensemble learning	98.12%	0.9577	Not provided	5000	5009
Bountakas and Xenakis (2023)	Ensemble learning	99.43%	0.9942	Not provided	32 051	3460
Proposed Model	Ensemble learning	**99.53%**	**0.9955**	**3.7**	5874	6220

resulting in a substantially lower classification accuracy not suitable for phishing detection, being such a high stakes issue.

While many existing models achieve high accuracy, they often rely on specific datasets or limited feature sets, making direct comparison challenging. For instance, models like THEMIS (Fang et al., 2019) and PhishNet-NLP (Verma et al., 2012) achieve impressive accuracy but utilise small, potentially biased datasets. Additionally, many studies focus on specific aspects of phishing detection, such as URL analysis (Sahingoz et al., 2019) or spear-phishing detection (Ding et al., 2021), limiting their generalisability to broader phishing threats. The proposed model addresses these limitations by leveraging a comprehensive set of features extracted from the email header, body, and URLs, providing a more holistic representation of phishing emails. The hybrid feature selection approach ensures the model captures a wide range of characteristics, improving its ability to detect diverse phishing techniques. Furthermore, the two-tiered stacking ensemble approach allows for the combination of predictions from multiple base learners, further enhancing accuracy and robustness.

There are few studies investigating the effectiveness of ensemble learning in phishing detection systems, as shown in Table 7. The proposed stacking ensemble learning model provides the highest accuracy when compared against recent and state-of-the-art comparable work from the literature. While (Bountakas and Xenakis, 2023) proposed a system that improved upon individual ML models, our work surpasses its accuracy and offers a simpler implementation. This research demonstrates the substantial benefit of incorporating ensemble learning techniques into phishing detection systems. The proposed stacking ensemble utilises a straightforward approach, relying on hybrid feature extraction without employing complex NLP techniques like (Bountakas and Xenakis, 2023). This simplicity contributes to the model's efficiency and makes it easier to implement and maintain. Moreover, the model leverages robust base learners (RF and XGBoost) both capable

of achieving over 99% accuracy individually. This choice of base learners, compared to the less effective KNN used in Bountakas and Xenakis (2023)' work, contributes significantly to the model's superior performance.

The proposed stacking ensemble addresses the limitations of traditional phishing detection methods, which often struggle to adapt to the evolving sophistication of phishing techniques. By combining the strengths of multiple algorithms and utilising a comprehensive set of features, the model effectively identifies and blocks malicious emails. The stacking ensemble's high accuracy and minimal impact on detection time make it a promising solution for real-world phishing detection systems. Its ability to effectively identify and block phishing emails can significantly reduce the risk of individuals and organisations falling victim to these attacks.

The proposed stacking ensemble learning model demonstrates strong performance. However, some limitations should be acknowledged. While this study has employed the recent and state-of-the-art datasets deployed in the recent literature, ensuring comparability with existing studies. We recognise that the restricted availability of comprehensive and up-to-date email datasets remains a significant challenge for all researchers in the phishing detection domain. For future work, we will focus on investigating the effectiveness of the proposed model on a more diverse dataset representative of the current phishing landscape and its evolving phishing techniques. It would be particularly valuable to expand the dataset to include a larger sample of emails, facilitating future testing of the model's scalability and its performance under stress conditions, including the processing of a continuous flow of email messages. Also, conducting experiments with both balanced and imbalanced datasets will offer valuable insights into the impact of data distribution on performance and help determine which approach is more effective when applying the model to unseen data. Additionally, incorporating email samples that contain attachment files would provide additional opportunities to enhance the accuracy of the models. By including this crucial element, a new hybrid feature type can be explored which extracts characteristics from email attachments, which often harbour malicious content.

While this study adopts a well-established hybrid feature extraction strategy, the broader landscape of phishing detection continues to evolve, presenting opportunities for further exploration. A comprehensive review of alternative feature extraction techniques, along with potential enhancements to the current approach, could contribute to more adaptive and robust models. Future work will investigate the impact of novel feature sets and advanced feature selection methods, which could offer an improved representation of emerging phishing tactics. Moreover, as phishing techniques are constantly evolving, reliance on a static and pre-collected dataset may not fully capture the dynamic nature of real-world scenarios. To address this, future research will focus on evaluating the proposed model in live, real-time environments, allowing for a more practical assessment of its effectiveness under realistic deployment conditions.

5. Conclusion

This paper aims to address the persistent problem of phishing, which remains one of the most prevalent initial access vectors in cyber attacks despite advancements in detection systems. Phishing poses significant risks to both individuals and organisations, necessitating continuous improvement in detection methodologies to counteract increasingly sophisticated attacks. Therefore, this paper proposed a novel stacking ensemble learning approach that leverages hybrid feature selection. The proposed model aims to enhance the effectiveness of phishing detection by combining predictions from multiple ML algorithms, each utilising different subsets of features extracted from various parts of the email, including the header, body, and URLs. The model leverages hybrid feature selection, combining diverse algorithms to accurately represent and identify phishing emails. The innovative approach involves a two-tiered stack, utilising strong base learners including RF and XGBoost, known for their high individual accuracy. The model's architecture allows for a comprehensive analysis of email content, incorporating lexical, syntactic, and URL features to capture a holistic understanding of phishing attempts.

The experimental results demonstrated that the proposed model achieved an accuracy of 99.53% and an F1-Measure of 0.9955, outperforming both individual ML algorithms and the most effective phishing detection systems documented in recent literature. The model's precision, recall, and AUC metrics were also high, indicating its robust performance in distinguishing between phishing and legitimate emails. By combining the strengths of multiple algorithms, the proposed model achieves higher accuracy and faster detection times compared to individual models. This approach not only enhances the detection capabilities but also simplifies the implementation process, making it more accessible for practical deployment in real-world scenarios. The model's ability to limit misclassification and its high AUC value further highlight its potential to significantly reduce the number of phishing emails reaching end users. For future work, we aim to enhance the proposed model by evaluating its performance on a broader dataset, refining the feature selection process, and assessing its scalability and efficiency in real-time applications.

CRediT authorship contribution statement

Aaron Connolly: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Hany F. Atlam:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Alhogail, A., Alsabih, A., 2021. Applying machine learning and natural language processing to detect phishing email. Comput. Secur. 110, 102414. http://dx.doi.org/10.1016/j.cose.2021.102414.

Alotaibi, R., Al-Turaiki, I., Alakeel, F., 2020. Mitigating Email Phishing Attacks using Convolutional Neural Networks. IEEE Xplore, pp. 1–6. http://dx.doi.org/10.1109/ICCAIS48893.2020.9096821, URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9096821.

Anandita Yadav, D.P., Paliwal, P., Kumar, D., Tripathi, R., 2017. A novel ensemble based identification of phishing E-mails. In: Proceedings of the 9th International Conference on Machine Learning and Computing. http://dx.doi.org/10.1145/3055635.3056583.

Apache, 2004. Index of /old/publiccorpus. spamassassin.apache.org, URL: https://spamassassin.apache.org/old/publiccorpus/.

Bountakas, P., Koutroumpouchos, K., Xenakis, C., 2021. A comparison of natural language processing and machine learning methods for phishing email detection. 16th Int. Conf. Availab. Reliab. Secur. http://dx.doi.org/10.1145/3465481.3469205.

Bountakas, P., Xenakis, C., 2023. HELPHED: Hybrid ensemble learning phishing email detection. J. Netw. Comput. Appl. 210, 103545. http://dx.doi.org/10.1016/j.jnca.2022.103545.

Che, H., Liu, Q., Zou, L., Yang, H., Zhou, D., Yu, F., 2017. A content-based phishing email detection method. pp. 415–422. http://dx.doi.org/10.1109/QRS-C.2017.75, URL: https://ieeexplore.ieee.org/document/8004352.

CMU, 2011. Enron Email Dataset. Cmu.edu, URL: https://www.cs.cmu.edu/~enron/.

Daeef, A., Badlishah Ahmad, R., Yacob, Y., Yaakob, N., Nurul, K., Ku Azir, F., 2016. Multi stage phishing email classification. J. Theor. Appl. Inf. Technol. 83.

Dewis, M., Viana, T., 2022. Phish responder: A hybrid machine learning approach to detect phishing and spam emails. Appl. Syst. Innov. 5, 73. http://dx.doi.org/10.3390/asi5040073.

Ding, X., Liu, B., Jiang, Z., Wang, Q., Xin, L., 2021. Spear phishing emails detection based on machine learning. In: 2021 IEEE 24th International Conference on Computer Supported Cooperative Work in Design. CSCWD, http://dx.doi.org/10.1109/cscwd49262.2021.9437758.

Dzuba, E., Cash, J., 2023. Introducing Cloudflare's 2023 Phishing Threats Report. The Cloudflare Blog, URL: https://blog.cloudflare.com/2023-phishing-report/.

Egozi, G., Verma, R., 2018. Phishing Email Detection Using Robust NLP Techniques | IEEE Conference Publication | IEEE Xplore. ieeexplore.ieee.org, URL: https://ieeexplore.ieee.org/document/8637476.

El Naqa, I., Murphy, M.J., 2015. What is machine learning? Mach. Learn. Radiat. Oncol. 1, 3–11. http://dx.doi.org/10.1007/978-3-319-18305-3_1, URL: https://link.springer.com/chapter/10.1007/978-3-319-18305-3_1.

Fang, Y., Zhang, C., Huang, C., Liu, L., Yang, Y., 2019. Phishing email detection using improved RCNN model with multilevel vectors and attention mechanism. IEEE Access 7, 56329–56340. http://dx.doi.org/10.1109/ACCESS.2019.2913705, URL: https://ieeexplore.ieee.org/document/8701426.

Gualberto, E.S., De Sousa, R.T., De B. Vieira, T.P., Da Costa, J.P.C.L., Duque, C.G., 2020. From feature engineering and topics models to enhanced prediction rates in phishing detection. IEEE Access 8, 76368–76385. http://dx.doi.org/10.1109/ACCESS.2020.2989126, URL: https://ieeexplore.ieee.org/abstract/document/9075252.

Gudivada, V., Apon, A., Ding, J., 2017. Data quality considerations for big data and machine learning: Going beyond data cleaning and transformations. URL: https://personales.upv.es/thinkmind/dl/journals/soft/soft_v10_n12_2017/soft_v10_n12_2017_1.pdf.

Hiransha, C.-D., Unnithan, N., Kp, S., 2018. Deep learning based phishing E-mail detection.

Hoo, Z.H., Candlish, J., Teare, D., 2017. What is an ROC curve? Emerg. Med. J. 34, 357–359. http://dx.doi.org/10.1136/emermed-2017-206735.

Jupyter, 2015. Jupyter notebook documentation — Jupyter notebook 7.1.0a2 documentation. URL: https://jupyter-notebook.readthedocs.io/en/latest/.

Karim, A., Shahroz, M., Mustofa, K., Belhaouari, S.B., Joga, S.R.K., 2023. Phishing detection system through hybrid machine learning based on URL. IEEE Access 11, 36805–36822. http://dx.doi.org/10.1109/access.2023.3252366.

Khonji, M., Iraqi, Y., Jones, A., 2013. Phishing detection: A literature survey. IEEE Commun. Surv. Tutor. 15, 2091–2121. http://dx.doi.org/10.1109/surv.2013.032213.00009.

Kumar, A., Chatterjee, J.M., Díaz, V.G., 2020. A novel hybrid approach of SVM combined with NLP and probabilistic neural network for email phishing. Int. J. Electr. Comput. Engineering (IJECE) 10, 486. http://dx.doi.org/10.11591/ijece.v10i1.pp486-493.

M., S., Pais, A.R., 2022. Classification of phishing email using word embedding and machine learning techniques. J. Cyber Secur. Mobil. http://dx.doi.org/10.13052/jcsm2245-1439.1131.

Ma, L., Ofoghi, B., Watters, P., Brown, S., 2009. Detecting phishing emails using hybrid features. In: 2009 Symposia and Workshops on Ubiquitous, Autonomic and Trusted Computing. http://dx.doi.org/10.1109/uic-atc.2009.103.

Matplotlib, 2024, Matplotlib: Python Plotting — Matplotlib 3.1.1 Documentation, Matplotlib.org, URL: https://matplotlib.org/.

Nazario, J., 2024. Index of / jose/phishing. monkey.org, URL: https://monkey.org/~jose/phishing/.

Nb, H., Kp, S., 2018. A machine learning approach towards phishing email detection CEN-security@iwspa 2018.

Nguyen, L.A.T., To, B.L., Nguyen, H.K., Nguyen, M.H., 2014. A Novel Approach for Phishing Detection Using URL-Based Heuristic. IEEE Xplore, pp. 298–303. http://dx.doi.org/10.1109/ComManTel.2014.6825621, URL: https://ieeexplore.ieee.org/document/6825621.

Niu, W., Zhang, X., Yang, G., Ma, Z., Zhuo, Z., 2017. Phishing Emails Detection Using CS-SVM. IEEE Xplore, pp. 1054–1059. http://dx.doi.org/10.1109/ISPA/IUCC.2017.00160, URL: https://ieeexplore.ieee.org/document/8367390.

Numpy, 2023. NumPy. Numpy.org, URL: https://numpy.org/.

Pandas, 2022. Python Data Analysis Library. Pydata.org, URL: https://pandas.pydata.org/.

Pavlyshenko, B., 2018. Using Stacking Approaches for Machine Learning Models. IEEE Xplore, pp. 255–258. http://dx.doi.org/10.1109/DSMP.2018.8478522, URL: https://ieeexplore.ieee.org/abstract/document/8478522.

Piepenbreier, N., 2023. One-hot encoding in machine learning with python •datagy. URL: https://datagy.io/one-hot-encoding/.

Python, 2024, Urllib — URL Handling Modules — Python 3.7.2 Documentation, Python.org, URL: https://docs.python.org/3/library/urllib.html.

Rahmi, I., Hamid, A., Abawajy, J., 2011. Hybrid feature selection for phishing email detection. LNCS 7017, 266–275.

Ramanathan, V., Wechsler, H., 2013. Phishing detection and impersonated entity discovery using conditional random field and latent Dirichlet allocation. Comput. Secur. 34, 123–139. http://dx.doi.org/10.1016/j.cose.2012.12.002.

Richardson, L., 2015. Beautiful Soup Documentation — Beautiful Soup 4.4.0 Documentation. URL: https://beautiful-soup-4.readthedocs.io/en/latest/.

Sahingoz, O.K., Buber, E., Demir, O., Diri, B., 2019. Machine learning based phishing detection from URLs. Expert Syst. Appl. 117, 345–357. http://dx.doi.org/10.1016/j.eswa.2018.09.029.

Salloum, S., Gaber, T., Vadera, S., Shaalan, K., 2021. Phishing email detection using natural language processing techniques: A literature survey. Procedia Comput. Sci. 189, 19–28. http://dx.doi.org/10.1016/j.procs.2021.05.077.

Scikit-learn, 2024, Scikit-Learn: Machine Learning in Python — Scikit-Learn 0.20.3 Documentation, Scikit-learn.org, URL: https://scikit-learn.org/stable/index.html.

seaborn, 2024, Seaborn: Statistical Data Visualization — Seaborn 0.9.0 Documentation, Pydata.org, URL: https://seaborn.pydata.org/.

Smadi, S., Aslam, N., Zhang, L., 2018. Detection of online phishing email using dynamic evolving neural network based on reinforcement learning. Decis. Support Syst. 107, 88–102. http://dx.doi.org/10.1016/j.dss.2018.01.001.

Smadi, S., Aslam, N., Zhang, L., Alasem, R., Hossain, M.A., 2015. Detection of Phishing Emails Using Data Mining Algorithms. IEEE Xplore, pp. 1–8. http://dx.doi.org/10.1109/SKIMA.2015.7399985, URL: https://ieeexplore.ieee.org/document/7399985.

Varoquaux, G., Colliot, O., 2023. Evaluating machine learning models and their diagnostic value. Neuromethods 601–630. http://dx.doi.org/10.1007/978-1-0716-3195-9_20.

Verma, R., Shashidhar, N., Hossain, N., 2012. Detecting phishing emails the natural language way.

XGBoost, 2022. Python package introduction — xgboost 1.5.2 documentation. URL: https://xgboost.readthedocs.io/en/stable/python/python_intro.html.

Yadav, N., Panda, S.P., 2021. Feature selection for email phishing detection using machine learning. Adv. Intell. Syst. Comput. 365–378. http://dx.doi.org/10.1007/978-981-16-2597-8_31.

Yang, Z., Qiao, C., Kan, W., Qiu, J., 2019. Phishing email detection based on hybrid features. IOP Conf. Ser. Earth Environ. Sci. 252, 042051. http://dx.doi.org/10.1088/1755-1315/252/4/042051.

Zhang, Z., Hamadi, H.A., Damiani, E., Yeun, C.Y., Taher, F., 2022. Explainable artificial intelligence applications in cyber security: State-of-the-art in research. IEEE Access 10, 93104–93139. http://dx.doi.org/10.1109/access.2022.3204051.

Aaron Connolly is a dedicated B.Sc. Cyber Security student at the University of Warwick, aspiring to become a PhD holder in the near future. With a solid background in security engineering from his industry experience, Aaron has ground his expertise in network security by applying his knowledge to practical, real-world situations. His research interests are deeply rooted in network and email security, particularly in developing innovative strategies to mitigate user-targeted cyber-attacks. Aaron is passionate about advancing preventative systems and enhancing user awareness to create a more secure digital environment. In addition to his academic pursuits, he actively participates in cybersecurity competitions and collaborates on research projects, demonstrating his commitment to staying at the forefront of the field.

Dr Hany Atlam is an assistant professor in Cyber security at WMG, University of Warwick, UK. He earned his PhD at the University of Southampton, UK in 2020. Prior to this, he completed his BEng (Hons) and MSc at Menoufia University, Egypt, in 2011 and 2014, respectively. Dr Atlam is a distinguished and highly cited researcher with more than 15 years of cutting-edge research and teaching experience in prestigious higher education institutions. He has authored and co-authored more than 50 research papers in high-ranked international journals and conferences.His significant contributions to the field have led to his recognition as one of the world's top 2 by Stanford University in three consecutive years, 2022, 2023 and 2024. Dr Atlam's research interests include and are not limited to Dynamic and Context-aware Access Control Models, IoT security and privacy, anomaly Intrusion detection, blockchain, risk-based access control, and digital forensics. Dr Atlam is key in providing innovative and effective security solutions to various domains. He is an active member of the research community, participating in peer review processes and serving on editorial boards of reputed journals.